Analytical Chemistry Cumulative Exam - December 1, 2023

Assigned paper: *Electrochemical Imaging of Interfaces in Energy Storage via Scanning Probe Methods: Techniques, Applications, and Prospects* doi:10.1146/annurev-anchem-091422-110703

Instructions: This exam is intended to assess cumulative knowledge and knowledge gained from reading the paper. The exam writer recognizes that this review article covers significant material and does not expect that you to have followed up on each reference. Each question gives you an opportunity to demonstrate your knowledge. You do not have to answer all questions correctly to pass.

Please numerically order your answers in your blue book from 1 to 9. That is, leave space if you skip one so you can return to it. To indicate the level of detail to provide in your answers the number of expected sentences is given – waffling answers containing extraneous details are strongly discouraged.

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1) The paper defines "anode" and "cathode" in *battery terminology*. What are these definitions? How do they differ from the definitions of anode and cathode taught in analytical chemistry courses (e.g., quantitative or instrumental analysis)? (2-3 sentences)

2) Is an SECM an *analytical instrument* or a *machine*? Justify your answer. (1-2 sentences)

3) Measurements of a battery material with SECM require a bipotentiostat. What is a bipotentiostat? and why do SECM measurements of batteries require one (2-3 sentences).

4) What is the SEI? (hint: be careful with your spelling of 'I' as there are two similar words with different meanings). What properties must the SEI have (in respect to ion and electron transport) in order to function as part of a battery (justify your answer)? (2-3 sentences)

5) The paper describes galvanostatic and potentiostatic cycling as two options for characterizing battery electrode. In each measurement, what quantity is controlled and what quantity is recorded? (1-2 sentences)

6) Of the four main techniques covered in the paper (EC-AFM, SECM, SECCM, and SICM) describe if the quantity that is recorded is due to a) an ionic current, b) faradaic current, c) other (please specify) – you do not need to justify your answer (1 sentence)

7) The 2nd reference in the assigned paper is an introductory review of batteries by M. Stanley Whittingham. What connects him to scientists Akira Yoshino and John B. Goodenough? (1-2 sentences)

8) Make a labelled drawing of the electrochemical set up for EC-AFM. You drawing should include all the appropriate electrodes (labelled), electrically connected to an appropriate instrument, which you may draw as a labelled box. (drawing, 0 sentences)

9) In one application of SECM described in the paper, the authors discuss work from their lab in which Hg was deposited onto a SECM microelectrode to make a Li^+ sensitive electrode. What electroanalytical technique does this suggest that the authors were using to do their detection? (1 sentence). Explain how this technique works (3-4 sentences)